

Supported degree-zero cohomology and exact arithmetic weights for Erdős–Straus split boxes

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Abstract

We instantiate the existing SplitZero internal quotient and original-Gram construction on the full labelled divisor boxes of an ES shell. A three-term finite cochain removes internal cancellations with explicit primitives; its degree-zero classes are exactly common channel/residue fibres. We compute their canonical metric and distinguish their dimension from both matched occurrences and original divisor-state counts. An exact inverse-fibre correction recovers every arithmetic mass and the entire original denominator-defect polynomial. Iterated quotients agree only when their inherited metrics are retained. All maps, inverses, empty fibres, coarsening kernels, and the original ES reconstruction are stated. No universal ES or RH claim or historical-priority conclusion is made.

1 Source interface: supported zero and original norms

The Zeta workbench’s SplitZero construction [1, 2, 3] is used here, not replaced by a second scalar theory. For a ring k , its coefficient semiring is $G(k) = \{\tau\} \sqcup k^\bullet$, with $e = 0_k^\bullet \neq \tau$. Reconstruction over a support semilattice L forms a disjoint union of module fibres. Its internal quotient is

$$(\lambda, x) \longmapsto (\lambda, [x]); \quad (\lambda, 0) \neq \tau \quad (\lambda \neq \perp).$$

The difference between two representatives is an actual relation, not a reason to remove the fibre label. In the source’s tau-base diagram the cochain is $[V \oplus V \xrightarrow{r_+ - r_-} \mathcal{B}]$ in degrees 0, 1. The analytic theta maps, their domains and $\mathcal{B}/\Theta V$ are not identified with a finite ES module in this note. We instantiate the source’s *algebraic* reconstruction, internal homology, kernel and original-Gram quotient interfaces on specified finite arithmetic maps. No purity, RH, or analytic-source comparison is imported.

The source’s least-norm construction is likewise retained. If $G > 0$ is the original Gram, B the independent relation columns and C a chosen representative matrix, then

$$R = C - B(B^*GB)^{-1}B^*GC, \quad R^*GR = C^*GC - C^*GB(B^*GB)^{-1}B^*GC. \quad (1)$$

The determinant of the source Gram in columns $[C, B]$ is $\det(B^*GB)\det(R^*GR)$. This is the existing Schur-complement calculation [3, §1], not a new determinant theorem. Our contribution below is a complete arithmetic instance, its counting weights and original selector inverse. The finite ES norm is specified directly; it is not silently declared equal to the theta-function norm.

2 The complete finite cochain and its quotient

Let X, Y, Z be finite sets with maps $\ell : X \rightarrow Z, r : Y \rightarrow Z$. For a commutative ring k , use the free modules with their labelled bases. Let $X_z = \ell^{-1}(z)$ and $Y_z = r^{-1}(z)$, and choose one reference x_z or y_z in each nonempty fibre. The relations are

$$B_X = \bigoplus_z \bigoplus_{x \in X_z \setminus \{x_z\}} k(e_x - e_{x_z}), \quad B_Y = \bigoplus_z \bigoplus_{y \in Y_z \setminus \{y_z\}} k(e_y - e_{y_z}).$$

Consider the actual three-term cochain

$$\mathcal{C}^{-1} = B_X \oplus B_Y \xrightarrow{\partial} \mathcal{C}^0 = k^X \oplus k^Y \xrightarrow{d} \mathcal{C}^1 = k^Z, \quad d(a, b)_z = \sum_{X_z} a_x - \sum_{Y_z} b_y. \quad (2)$$

The incoming map is inclusion; $d\partial = 0$ term by term.

Theorem 1 (Complete integral cohomology and inverse). *Put $S = \ell(X)$, $T = r(Y)$ and $I = S \cap T$. Over every commutative ring,*

$$H^{-1}(\mathcal{C}) = 0, \quad H^0(\mathcal{C}) \simeq k^I, \quad H^1(\mathcal{C}) \simeq k^{Z \setminus (S \cup T)}. \quad (3)$$

The degree-zero isomorphism sends a cycle to $c_z = \sum_{X_z} a_x = \sum_{Y_z} b_y$. Its inverse sends c_z to $c_z(e_{x_z}, e_{y_z})$. The complete fibre over (c_z) consists of this representative plus $\partial\theta$, with unique relation coefficients $\theta_x = a_x$ and $\theta_y = b_y$ away from the reference indices.

Proof. The displayed star differences are independent and generate exactly the kernel of each fibre sum: a vector of sum zero equals $\sum_{x \neq x_z} a_x(e_x - e_{x_z})$. In a cycle the two sums agree. If only one fibre is nonempty, this common sum is zero; otherwise it is any c_z . Subtracting the stated reference representative leaves precisely the displayed relations. This proves (3), including both inverse compositions, without division in k . Finally the image of d contains exactly all coordinate basis vectors indexed by $S \cup T$. $\square$

Taking only $\ker d$ without the original incoming relations would count internal cancellations as witnesses. The relation primitive in the proof specifies exactly what is removed. To apply the source's support theory, use the four masks $L = \mathcal{P}(\{\text{left}, \text{right}\})$, coordinate inclusions at present legs, and the common codomain for every nonempty mask. Set the empty-mask complex to zero. The relation submodules are transition-stable. Reconstruct (2) and its homology internally as in [1, §§1–4]; the differential-square equation is the supported-zero map at the retained mask, not the constant absent map. If the joint H^0 is zero, that is still its *present zero fibre*.

2.1 The complete canonical weight

Give $k^X \oplus k^Y$ its original real or complex diagonal Gram, with positive entries a_x, b_y . Define

$$M_z = \sum_{X_z} a_x^{-1}, \quad N_z = \sum_{Y_z} b_y^{-1}.$$

For $z \in I$, its least-norm representative is

$$r(c)_x = \frac{c_z}{a_x M_z}, \quad r(c)_y = \frac{c_z}{b_y N_z}, \quad (4)$$

and its quotient norm is

$$\|c\|_{H^0}^2 = \sum_{z \in I} (M_z^{-1} + N_z^{-1}) |c_z|^2. \quad (5)$$

To prove this, the weighted inner product of (4) with each star relation is zero. Its sums are c_z , and its squared norm is (5). Subtracting it from any other cycle supplies the explicit relation primitive and the Pythagorean identity. Empty fibres are excluded from (4), not assigned a fictitious inverse weight. Least-norm representatives may be rational or complex; integral witness extraction uses labelled occurrences, not the claim that (4) is an integral source vector.

For the original counting Gram $a_x = b_y = 1$, put $m_z = |X_z|$, $n_z = |Y_z|$. Then the quotient Gram is $\text{diag}(1/m_z + 1/n_z)$. In the integral cycle basis given by the reference representative and star differences, a common fibre has source determinant $m_z + n_z$, relation determinant $m_z n_z$, and quotient determinant $(m_z + n_z)/(m_z n_z)$. A one-sided fibre contributes the same determinant to

source and relations. These formulas follow either from (1) or from $\det(I + \mathbf{1}\mathbf{1}^*) = \text{matrix size} + 1$ on each star block. Thus all source and relation factors are visible.

The canonical projection of the original all-ones vector onto the harmonic degree-zero subspace has components $2n_z/(m_z + n_z)$ on X_z and $2m_z/(m_z + n_z)$ on Y_z , with squared norm

$$\boxed{4 \sum_{z \in I} \frac{m_z n_z}{m_z + n_z}}. \quad (6)$$

This is positive exactly when I is nonempty. It is an exact source-norm criterion, not a bound proving I nonempty for arbitrary arithmetic data.

2.2 Coarsening and killed classes

For $\pi : Z \rightarrow Z'$, retain the same X, Y but replace their maps by $\pi\ell, \pi r$. The identity on degree zero and fibre-sum map on degree one define a chain map; the incoming map sends each old star difference to its explicit expression in the coarser star basis. The induced map is

$$(c_z)_{z \in I} \mapsto \left(\sum_{\substack{z \in I \\ \pi(z)=w}} c_z \right)_{w \in I'}, \quad I' = \pi(S) \cap \pi(T). \quad (7)$$

Its image is $k^{\pi(I)}$, its cokernel $k^{I' \setminus \pi(I)}$, and its kernel has the star-difference basis in each fibre of $I \rightarrow \pi(I)$. For any killed class its chain representative has coarser sums zero, and the preceding nonreference-coordinate formula is its boundary primitive. This is an actual instance of the source's transported-class kernel, not the assumption that comparisons are injective [1, §3]. For example two matched residues merging to one kill the nonzero class $(1, -1)$; unmatched fine residues can also create new coarse matches.

3 The ES specialization: two channels, all actual exponents

In this arithmetic application, the coefficient field is $\mathbb{Q}$. Fix a prime $p \equiv 1 \pmod{4}$, $p/4 < a < p/2$, $R = 4a - p$, and $a = \prod_i q_i^{e_i}$. Put $B_e = \prod_i [-e_i, e_i]_{\mathbb{Z}}$ and $v(\beta) = \prod_i q_i^{\beta_i} \pmod{R}$. All powers are units. Retain the two *canonical* channel targets

$$t_M = -1, \quad t_E = -p^{-1} \pmod{R}. \quad (8)$$

The additional inverted-exterior target of the split-energy test is a labelled alternative computation of E , not a third canonical channel.

Split $e = f + b$ with $f, b \geq 0$. In each separately labelled channel use

$$X = B_f, \quad Y = B_b, \quad Z = (\mathbb{Z}/R\mathbb{Z})^\times, \quad \ell(\xi) = v(\xi), \quad r_c(\eta) = t_c v(\eta)^{-1}. \quad (9)$$

The full object is the direct sum of these channel complexes, retaining (p, a, R, c, f, b) as its type. Theorem 1 proves

$$H^0(\mathcal{C}_c) \neq 0 \iff \exists(\xi, \eta) : v(\xi + \eta) = t_c. \quad (10)$$

Every $\beta \in B_e$ has the complete addition fibre

$$\max(-f_i, \beta_i - b_i) \leq \xi_i \leq \min(f_i, \beta_i + b_i), \quad \eta = \beta - \xi, \quad (11)$$

so (10) is equivalent to an original channel hit. The original divisor is

$$u = \prod_i q_i^{e_i + \beta_i}, \quad d_0 = \gcd(a, u), \quad (h, r, s) = (d_0^2/u, u/d_0, a/d_0). \quad (12)$$

On M use $\lambda = (r + s)/R$ and denominators $(a, phs\lambda, phr\lambda)$; on E use $\kappa = (pr + s)/R$ and $(a, hsk, phr\kappa)$. The inverse is $u = pa^2/(RY - pa)$ on M and $u = a^2/(RY - pa)$ on E. Prime valuations prove all normalizations; clearing the reciprocal sum proves the original ES identity. These coordinates and first-half completeness are inherited [6, 5, 4]. The ordered pair (ξ, η) , channel and full exponents are not inferred from a residue alone; they are selected from the recorded fibres (9),(11).

4 The exact original arithmetic weight and its chain primitive

Let $\mathcal{P}_c$ be the matched pairs in (9) and let $\mathcal{W}_c = \{\beta \in B_e : v(\beta) = t_c\}$. Addition gives a surjection

$$q : \mathcal{P}_c \rightarrow \mathcal{W}_c, \quad N(\beta) = |q^{-1}(\beta)| = \prod_i (\min(f_i, \beta_i + b_i) - \max(-f_i, \beta_i - b_i) + 1). \quad (13)$$

There are *two* observations of $\mathcal{P}_c$: its matched residue and its original exponent vector. Neither determines the other in general. The residue cohomology rank, the matching-pair count and the original witness count must not be identified.

For any finite surjection $q : P \rightarrow W$, let $Q : k^P \rightarrow k^W$ sum over its fibres. Its kernel has the complete basis $e_x - e_{x_0(w)}$. Thus

$$[\ker Q \hookrightarrow k^P] \simeq k^W[0] \quad (14)$$

in degrees $-1, 0$, by the integral reference section and explicit fibre relations. Over $\mathbb{Q}$ the canonical section for the counting Gram is

$$J(c)_x = c_{q(x)}/N(q(x)), \quad QJ = 1. \quad (15)$$

For every amplitude vector A , retain both QA and $\theta_x = A_x - J(QA)_x$ at each nonreference point. This is a linear bijection, with inverse $J(QA) + \sum \theta_x(e_x - e_{x_0})$. In particular a nonzero cancellation is an actual boundary with its primitive, not external absence.

The inherited counting norm gives

$$\|A\|^2 = \sum_w \frac{|QA(w)|^2}{N(w)} + \|A - JQA\|^2. \quad (16)$$

If instead one is transporting the *original uniform norm* on k^W , its permutation-invariant diagonal metric on a refined fibre is necessarily $N(w)I$: minimization then gives quotient norm $|c_w|^2$. This transported metric is displayed as a changed metric; it is not called the native product-counting norm. Indeed a constant diagonal $h_w I$ on a fibre has quotient norm $h_w |c_w|^2/N(w)$, proving uniqueness.

Over $\mathbb{Q}$ or $\mathbb{R}$, arbitrary positive original ES masses $\omega_{c,\beta}$ are reconstructed without duplication by assigning to each lifted pair the mass $\omega_{c,\beta}/N(\beta)$. In particular the full original channel count and every monomial are recovered exactly:

$$\boxed{\sum_{\beta \in \mathcal{W}_c} U^{e+\beta} = \sum_{(\xi, \eta) \in \mathcal{P}_c} \frac{U^{e+\xi+\eta}}{N(\xi + \eta)}}. \quad (17)$$

The zero coefficient is absence of that monomial; a signed intermediate sum cancelling to zero retains its source support mask separately.

4.1 All failed candidates retain their original defect

For every split pair, matched or not, put

$$x = v(\xi), \quad w = v(\eta), \quad \rho = x - t_c w^{-1} \pmod{R}, \quad \beta = \xi + \eta.$$

The original gates $g_M = u + a$, $g_E = 4u + 1$ satisfy

$$g_c \equiv c_c \rho \pmod{R}, \quad c_M = aw, \quad c_E = pw, \quad (18)$$

where both multipliers are units. Hence

$$D_c(\beta) = \frac{R}{\gcd(R, g_c)} = \frac{R}{\gcd(R, \rho)}. \quad (19)$$

For the original affine monodromy

$$(X, Y, Z) \mapsto (6g + Y, -6g - X - Y, Z - 2g + X), \quad (-Y, X - 6g, Z + 3g + Y), \quad (X, Y + X, Z - g),$$

scaling by c_c is an equivariant bijection from coefficient ρ to coefficient g_c , inverse scaling by c_c^{-1} . No global sphere theorem is required. Equations (13),(18) prove the *complete defect-polynomial* identity

$$\sum_{\beta \in B_c} U^{e+\beta} Z^{D_c(\beta)-1} = \sum_{\xi \in B_f, \eta \in B_b} \frac{U^{e+\xi+\eta} Z^{D_c(\xi+\eta)-1}}{N(\xi + \eta)}. \quad (20)$$

Evaluating the polynomial at $Z = 0$ means coefficient extraction at exponent zero and gives (17). This is not a limiting positivity assumption.

4.2 Iterated quotients must carry their inherited metric

For finite surjections $P \xrightarrow{q} W \xrightarrow{r} Z$, let $m_w = |q^{-1}(w)|$ and $n_z = \sum_{r(w)=z} m_w$. The quotient metric on k^W is $\text{diag}(1/m_w)$. The canonical lift of c_z to w is $m_w c_z / n_z$; lifting again gives c_z / n_z at every original occurrence. It agrees exactly with the direct canonical lift for rq . Every kernel coordinate is recovered by the successive star primitives. This proves all finite towers by induction.

Resetting the intermediate metric to the identity changes that map. For three budgets $[-1, 1]$, total exponent zero has seven ordered triples. The direct coefficients are all $1/7$ and squared norm $1/7$. Resetting after adding the first two gives four coefficients $1/6$ and three $1/9$, with squared norm $4/27$. Thus the exact excess is

$$\boxed{\frac{4}{27} - \frac{1}{7} = \frac{1}{189} > 0.} \quad (21)$$

The discarded difference is a nonzero zero-sum boundary. This is a concrete reason to retain the source/relation metric through each morphism.

5 Concrete tests and the counterfactual object

At $p = 2521$, $a = 636 = 2^2 \cdot 3 \cdot 53$, $R = 23$, choose $f = (1, 1, 1)$, $b = (1, 0, 0)$. The complete matched pair table is

c	g	ξ	η	β	$N(\beta)$
M	22	$(-1, 1, 1)$	$(0, 0, 0)$	$(-1, 1, 1)$	2
M	11	$(0, -1, -1)$	$(1, 0, 0)$	$(1, -1, -1)$	2
M	21	$(0, 1, 1)$	$(-1, 0, 0)$	$(-1, 1, 1)$	2
M	22	$(1, -1, -1)$	$(0, 0, 0)$	$(1, -1, -1)$	2
E	13	$(-1, 1, 0)$	$(-1, 0, 0)$	$(-2, 1, 0)$	1

There are five matched pairs, four channel/residue cohomology coordinates, and three original canonical channel states. The harmonic Gram in the ordered shared coordinates

$$(M, 11), (M, 21), (M, 22), (E, 13)$$

is $\text{diag}(2, 2, 3/2, 2)$, determinant 12. The mass identity is $4 \cdot (1/2) + 1 = 3$, not five. The two middle states have $(u, h, r, s, \lambda) = (8, 2, 2, 159, 7)$ and $(50562, 2, 159, 2, 7)$; the exterior state has $(477, 53, 3, 4, 329)$. Equation (12) returns all denominators and (17) retains the two middle orientations as distinct canonical labels.

At the genuinely empty *single shell* $p = 241, a = 64, R = 15$, split $6 = 3 + 3$. The two channel complexes together have dimensions

$$\dim C^{-1} = 12, \quad \dim C^0 = 28, \quad \dim C^1 = 16, \quad H^{-1} = H^0 = H^1 = 0.$$

The raw two-leg kernel has dimension twelve; all twelve coordinates are actual relations. The counting norm of the all-ones source vector is 28, not zero. Its supported acyclic fibre is not τ . Other shells at 241 succeed, so this is not an ES counterexample.

Likewise at $p = 37, a = 18, R = 35$, the exterior local divisor sets are $\{1, 6, 36, 81\}$ modulo five and $\{12, 54\}$ modulo seven. They are both nonempty but disjoint. Equation (7) permits new cohomology after coarsening; it does not give a simultaneous original exponent vector. One must retain the joint residue or the same original leaf in every local factor. These original regression sets are inherited [5].

For the complete first-half atlas at a fixed prime, choose any split at each shell and sum the two canonical channel complexes with their full support labels. Its degree-zero cohomology is nonzero exactly when ES holds at that prime. A hypothetical false prime has zero ordinary H^0 in every such labelled summand, not absent source objects or zero source weights. On the refined matched-pair complexes (14), H^0 counts actual canonical states; on (2) it counts matched channel/residues. The distinction persists under all split changes.

These identities do not produce a universal nonzero class. They supply an exact source-norm expression (6), a complete original state-count expression (17), and an all-defect expression (20) on which a positivity or cross-shell estimate can be attempted. A generic source-norm bound from the Zeta application is not presumed valid for this finite arithmetic family merely because the quotient formulas have the same algebraic form.

6 Verification scope

The standard-library verifier constructs the actual differentials, relation columns and canonical representatives. It checks their ranks, both inverse compositions, relation primitives, and determinant ratios for every pair of maps from sets of size at most two into a three-element set, in both counting and specified nonconstant diagonal metrics. It also checks unequal common fibres through size six, every small split budget, and the complete three-stage tower. The displayed arithmetic examples include every original candidate and every split pair, successful or failed. Further bounded first-half scans retain their exact domain in the JSON. They prove no unbounded occupancy statement.

Source-level general statements and finite checks have different scopes. The Zeta analytic source is read through the pinned workbenches cited below; the full 482-page reader and the analytic/formal corpus were not audited or replayed. No Lean theorem is claimed for this tranche. Existing mathematical source files are not rewritten, and no historical-priority claim is made.

References

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- [6] Elsholtz, C., & Tao, T. (2013). Counting the number of solutions to the Erdős–Straus equation on unit fractions. *Journal of the Australian Mathematical Society*, 94(1), 50–105. Section 2, Propositions 2.2 and 2.6. <https://arxiv.org/abs/1107.1010>.